

**Defensive Efficiency in Path of Exile 2:
Physical Damage Mitigation
Across the Ascendancy Spectrum**

Shaman, Infernalist, Witchhunter, Tactician, Warbringer, Invoker, and the Smith of

Kitava Benchmark

Path of Exile 2 — Defensive Layer Theory

July 2026

Contents

1	Conversion Stack and Sources	5
2	Mathematical Framework and Build Parameters	6
3	Hit-Size Scaling Analysis	7
4	Smith of Kitava vs. Shaman: The Divergence Narrative	9
5	The 2,000 HP Benchmark and Survival Thresholds	12
6	Armour Efficiency Amplification	14
7	Full Defensive Layer Synergy	15
8	Passive Point Economy	17
9	Progressive Investment Breakpoints	18
10	Infernalist: Chaos Inoculation and the 20% Hard Immunity	19
11	Invoker: Avoidance, Evasion-as-Armour, and the Conditional eHP Ceiling	25
12	Witchhunter: Sorcery Ward and the Absorption-Layer Paradigm	35
13	Tactician: Deflection, Capped DR, and the Armour-Efficient Paradigm	40
14	Warbringer: Block-as-Mitigation and the Mapping Specialist	47
15	Conclusion	53

Executive Summary

- **Conversion Stack (Corrected):** The Shaman converts 75% of incoming physical damage to elemental — 60% to fire (Cloak of Flame 50% + ascendancy 5% + passive tree notable 5%), 5% each to cold, lightning, and a random element. Only 25% remains physical. Prior analyses cited 60–70%; the additional passive tree notable raises the total to 75%.
- **Both Archetypes Look Comparable at Small Hits.** Against a 100-damage hit, a Smith of Kitava with 100,000 armour achieves 99.0% mitigation and a Shaman with 25,000 armour and 75% conversion achieves 99.5%. The 0.5-percentage-point gap is meaningless in practice — armour's formula is overwhelmingly efficient for any build at this scale, and the Shaman's conversion adds only fractional benefit when the raw hit is already small.
- **Divergence is Continuous and Compounds with Hit Size.** From 1,000 to 10,000 incoming damage the gap grows from 5 to 30.8 percentage points. Smith of Kitava's 100,000 armour falls from 90.9% to 50.0% mitigation over this range. Shaman's 25,000 armour with conversion falls only from 95.9% to 80.8% — the conversion suppresses the effective physical hit size at every point, keeping the armour formula in its efficient operating range.
- **The 2,000 HP Survival Threshold.** At $H=10,000$ (the practical endgame hit ceiling), Shaman takes 1,923 damage. A 2,000 effective HP pool is sufficient to survive the largest hit in the game with 77 HP remaining. Smith takes 5,000 from the same hit and requires 5,001 effective HP to survive — a $2.6\times$ higher life requirement, demanding entirely different investment priorities.
- **The Advantage Cannot Be Closed Through Armour Alone.** To match Shaman's total mitigation via armour with no conversion requires approximately 421,000 armour rating. Smith's practical ceiling of 100,000 achieves 50% at the endgame hit ceiling versus Shaman's 80.8% — a structural gap, not an incremental one.
- **Passive Point Economy.** Shaman's entire conversion stack costs one passive tree notable, a few ascendancy notables, and Cloak of Flame. The 8–15 passive points Smith commits to armour scaling are freed for damage, sustain, or additional defensive layers — enabling Shaman to simultaneously out-defend a dedicated armour archetype and retain budget flexibility.

- **Infernalist: Chaos Inoculation and the Hard Immunity Advantage.** The Infernalist (Witch ascendancy) achieves comparable or superior mitigation through a distinct conversion stack: 50–55% physical-to-fire, 20% physical-to-chaos eliminated entirely via Chaos Inoculation, and a reduced physical residual of 20–30%. The CI keystone makes the chaos component a categorical zero rather than a near-zero residual, and simultaneously frees all chaos resistance affixes and life affixes on gear for Energy Shield and evasion. The Infernalist Full configuration (80% total conversion + CI) achieves 85.3% mitigation at $H=10,000$ — surpassing standard Shaman at 80.8% — while providing hard chaos immunity at every hit size.
- **Warbringer: Block-as-Mitigation and the Mapping Specialist.** The Warbringer (Marauder ascendancy) converts block into a front-loaded damage multiplier via Turtle Charm: blocked hits deal 20% of their original value to the player (80% less), raising the effective block cap to 75%. Wooden Wall adds a second front-loaded layer — 20% of damage taken goes to an active totem instead. Combined on a blocked hit: $0.20 \times 0.80 = 0.16H$ enters the armour formula. At $A=100k$, the expected mitigation across the probability chain (75% block) is 89.5% — the highest of any armour-anchored archetype in this paper. The binding constraint is the 25% of unblocked hits, which arrive at 80% of their original value and impose a 3,557 HP minimum survival requirement at $H=10,000$. The build excels in mapping content where the block distribution averages out; it is exposed to single-hit variance in boss encounters, partially offset by Turtle Charm’s unique ability to block otherwise unblockable hits.
- **Tactician: Deflection, Capped DR, and High Variance.** The Tactician (Mercenary ascendancy) achieves armour efficiency through “Stay Light, Use Cover,” which defends with 200% of armour — halving the armour required to reach any given DR — while hard-capping both evasion and damage reduction at 50%. “Polish that Gear” converts 20% of armour into deflection rating; deflection reduces a hit by 40% before armour evaluates. A deflected hit at the 50% DR cap takes 30% of its original value (70% effective DR), comparable to Shaman. An undeflected hit at the cap takes 50%, requiring 5,001 effective HP to survive a 10,000 raw hit — identical to Smith of Kitava’s requirement. Expected mitigation across the probability chain (25% effective evasion, 75% deflection with Glancing Blows) is 73.8% at $H=10,000$, trailing Shaman’s deterministic 80.8%. The build is efficient when deflection fires; it is exposed when it does not.
- **Witchhunter: Sorcery Ward and the Absorption-Layer Ceiling.** The

Witchhunter (Mercenary ascendancy) achieves the highest single-hit survival ceiling of the archetypes examined in this paper. Two ascendancy notables (Obsessive Rituals + Ceremonial Ablution) convert armour and evasion investment into a Sorcery Ward absorption pool ($SW = 30\%$ of post-penalty armour+evasion, capped at 32,000) that intercepts all post-armour hit damage before it reaches life. At 60,000 geared armour (30,000 post-penalty) and $L = 3,000$ HP, $SW = 9,000$ — absorbing the full post-armour remainder of a 10,000 raw hit with zero life damage. The derived survival ceiling is approximately $H = 14,485$, substantially exceeding both Shaman and Smith at comparable passive investment. No unique item is required.

- **Invoker: Avoidance-Primary Architecture and Conditional eHP Ceiling.** The Invoker (Monk ascendancy) achieves defensive efficiency through a fundamentally different paradigm: preventing hits from connecting via evasion and block, rather than suppressing damage on hits that land. The “...and protect me from harm” ascendancy node unifies the evasion and armour pools, applying 65% of base evasion as effective armour, so that investment in evasion simultaneously improves avoidance chance and provides armour DR. At 50%+50% evasion and block, 75% of all hits are avoided before any mitigation layer applies. Against the hits that do land, Meditate — a pre-combat skill that overflows Energy Shield to $1.5\times$ — provides the eHP buffer. The result is an archetype with lower per-hit DR than Shaman (27–41% at $H=10,000$) but dramatically lower total encounter damage intake under optimal play.

1 Conversion Stack and Sources

The Shaman’s physical-to-elemental conversion assembles from six distinct sources: a unique chest item, three ascendancy notables, a passive tree notable, and a crafted anointment. Each source stacks additively.

Table 1: **Shaman Physical-to-Elemental Conversion Stack — All Sources**

Source	Type	Target Element	Conversion
Cloak of Flame	Unique Chest Item	Fire	50%
Ascendancy Notable	Ascendancy	Fire	5%
Passive Tree Notable	Passive Skill Node	Fire	5%
Ascendancy Notable	Ascendancy	Cold	5%
Ascendancy Notable	Ascendancy	Lightning	5%
Anointment	Ring / Amulet Craft	Random Element	5%
Total Elemental Converted			75%
Residual Physical			25%

Note: Fire receives 60% in total (50 + 5 + 5). Cold, lightning, and the random anointment element each receive 5%. The bolded passive tree notable (5% physical taken as fire) is the corrected source that prior analyses omitted, raising the aggregate from 60–70% to 75% and changing residual physical from 40% to 25%.

The correction from prior analysis matters nonlinearly. Moving from 60% to 75% conversion reduces residual physical from 40% to 25% of the original hit — a 37.5% reduction in the physical component that the armour formula must handle. Because armour’s efficiency scales with *how small* the incoming hit is (not linearly with conversion), this final increment of conversion produces disproportionately large gains, as the hit-size tables in Section 3 will quantify.

2 Mathematical Framework and Build Parameters

The Path of Exile 2 Armour Formula

$$\text{DR}_{\text{phys}} = \frac{A}{A + 10H}$$

where A is armour rating and H is the size of the incoming physical hit. The formula exhibits strongly diminishing returns: the denominator grows linearly with hit size, causing DR% to decay increasingly fast as H increases. This is armour's fatal structural weakness against large hits. The only way to counteract it without increasing A is to reduce H — which is precisely what conversion does.

When armour applies to elemental damage (“X% of Armour applies to Elemental Damage”), the same formula evaluates each elemental hit component independently:

$$\text{DR}_{\text{ele}} = \frac{A}{A + 10 \cdot H_{\text{ele}}}$$

At 100% armour-to-elemental (achievable by Shaman through ascendancy nodes), the full armour value applies. Critically, the cold, lightning, and random elemental components are each only 5% of the raw hit — for a 10,000 raw hit, each is only 500 damage. At 25,000 armour:

$$\text{DR vs 500 ele hit} = \frac{25,000}{25,000 + 5,000} = 83.3\%$$

These small components are nearly eliminated by armour alone, before resistance even applies. Only the 60% fire component (6,000 at $H=10,000$) represents a meaningful elemental contribution.

Build Parameters

Three archetypes are compared throughout this analysis. Shaman and Baseline share identical armour values — all Shaman mitigation improvement relative to Baseline is from conversion and elemental layers, not armour scaling.

Table 2: **Build Parameters for Cross-Archetype Comparison**

Parameter	Baseline	Smith of Kitava	Shaman
Armour Rating	25,000	100,000	25,000
Conversion (phys $\rightarrow$ ele)	0%	0%	75%
Armour-to-Elemental	0%	0%	100%
Max Elemental Resistance	75%	80%	85%
Armour investment	1$\times$	4$\times$	1$\times$

Note: Smith of Kitava’s 100,000 armour represents heavy ascendancy and passive investment — a realistic ceiling for a dedicated armour archetype. Shaman uses identical base armour to the Baseline build; all mitigation improvements come from conversion. Resistance differences reflect Shaman’s access to the Primal Aegis ascendancy path and Smith’s strong but less extreme resistance ceiling. Incoming hit is treated as 100% physical in all cases; any elemental component exists only as converted damage within the Shaman’s defensive stack.

3 Hit-Size Scaling Analysis

This section is the core analytical output: how total effective mitigation and absolute damage taken changes across every meaningful hit size in Path of Exile 2 content, from minor chip damage (100) through the practical one-shot threshold (10,000). All Shaman calculations use the per-element formula, which independently applies armour to each converted component and captures the near-elimination of the small cold/lightning/random portions.

Effective Mitigation by Hit Size

The colour gradient tells the full story. Reading Smith’s column downward: four hit sizes in green (near-perfect), then yellow, orange, and two in red at the endgame ceiling. Reading Shaman’s column: three hit sizes in dark green, one in light green, and the final three in yellow — the conversion keeps the build in the efficient range of the armour formula at every point on the spectrum.

Two observations stand out:

1. **At H=100, both are green** — the columns look practically identical (99.0% vs 99.5%). This confirms the user intuition that armour “performs exceptionally well” at low hit values regardless of build; the conversion adds almost no visible benefit here.
2. **At H=10,000, the columns are different tiers** (red vs yellow) with a 30.8

Table 3: Total Effective Mitigation (%) by Incoming Hit Size

Incoming Hit	Baseline (25k)	Smith of Kitava (100k)	Shaman (25k + 75%)	Smith → Sha
100	96.2%	99.0%	99.5%	
500	83.3%	95.2%	97.8%	
1,000	71.4%	90.9%	95.9%	
2,000	55.6%	83.3%	92.8%	
4,000	38.5%	71.4%	88.3%	
8,000	23.8%	55.6%	82.7%	
10,000	20.0%	50.0%	80.8%	

Cell shading by mitigation tier: dark green $\geq 95\%$, light green 90–94%, yellow 80–89%, orange 70–79%, red $< 70\%$. “Gap” is Shaman mitigation minus Smith mitigation in percentage points (pp). Per-element armour formula used for Shaman throughout; cold/lightning/random components (5% of raw hit each) are nearly eliminated by armour before resistance applies.

pp gap — the same investment that produced negligible difference at small hits is now the difference between taking 50% of a hit versus 19.2%.

Absolute Damage Taken by Hit Size

At small hit sizes, both the damage-taken numbers and the minimum HP requirements are negligible for both builds — a character with any meaningful life pool survives regardless. At the endgame ceiling, the minimum HP requirement diverges by more than 3,000: Shaman needs 1,923 HP, Smith needs 5,001 HP to guarantee survival against an identical raw hit.

Table 4: **Damage Taken (Absolute Values) by Incoming Hit Size**

Hit	Baseline	Smith	Shaman	Min HP: Smith	Min HP: Shaman
100	4	1	<1	1	<1
500	83	24	11	24	11
1,000	286	91	41	91	41
2,000	889	333	143	333	143
4,000	2,462	1,143	469	1,143	469
8,000	6,095	3,556	1,387	3,556	1,387
10,000	8,000	5,000	1,923	5,001	1,923

“Min HP” is the minimum effective HP pool needed to survive a single hit of the listed size without dying (damage taken + 1). Baseline and Smith receive 100% physical; no resistance applies. Shaman values use per-element calculation. All values rounded to nearest whole number. The Shaman / Smith ratio on taken damage grows from $2.1\times$ at $H=100$ to $2.6\times$ at $H=10,000$.

4 Smith of Kitava vs. Shaman: The Divergence Narrative

Why Both Look Comparable at Small Hits

At hit sizes of 100–500, the armour formula’s denominator is dominated by the armour value A :

$$H = 100: \quad \frac{100,000}{100,000 + 1,000} = 99.0\% \quad \frac{25,000}{25,000 + 250} = 99.0\%$$

When $A \gg 10H$, the formula approaches 100% for any build with substantial armour. A Baseline build with only 25,000 armour already reaches 96.2% at $H=100$. Smith and Shaman both sit in the 99% range because the denominator addition ($10H = 1,000$) is negligible compared to A . Conversion adds fractional benefit: 0.46 damage vs. 1.0 damage. Both are effectively zero in context.

This is the operating regime where armour is “exceptionally well” — and it is the regime that blinds pure-armour thinking to the problem. The armour formula feels infallible at small hits.

The Break Point: Where Hit Size Threatens the Formula

As H grows past the 1,000–2,000 range, the term $10H$ becomes a meaningful fraction of A . The formula degrades:

$$\begin{aligned} H = 1,000: \quad \text{Smith} &= \frac{100,000}{110,000} = 90.9\% \\ H = 4,000: \quad \text{Smith} &= \frac{100,000}{140,000} = 71.4\% \\ H = 10,000: \quad \text{Smith} &= \frac{100,000}{200,000} = 50.0\% \end{aligned}$$

Smith of Kitava has no mechanism to suppress H . Every additional 1,000 damage of incoming hit adds 10,000 to the denominator, and the formula's efficiency erodes in direct proportion. There is no passive investment that changes this — armour can only scale A , and A must grow faster than $10H$ to maintain DR, which requires exponentially more armour as hits scale.

Shaman's conversion intercepts this degradation by *permanently suppressing the value of H the formula evaluates against*. A 10,000 raw physical hit arrives at the armour formula as:

- **Physical component:** 2,500 effective hit — the formula sees a $4\times$ smaller hit
- **Fire component (6,000):** handled by armour-to-elemental + 85% resistance
- **Cold/lightning/random (500 each):** nearly eliminated (83.3% DR from armour alone)

The same 25,000 armour that gives 20% DR against a 10,000 physical hit gives 50% DR against the 2,500 physical component — matching what Smith achieves with 100,000 armour on the full physical hit:

$$\text{Shaman vs 2,500: } \frac{25,000}{50,000} = 50.0\% \quad = \quad \text{Smith vs 10,000: } \frac{100,000}{200,000} = 50.0\%$$

The physical DR is identical. Then Shaman also eliminates 75% of the hit through elemental layers. Smith has no equivalent layer.

The “Below 90%” Threshold

A useful concrete benchmark is when each build’s mitigation falls below 90% — a level at which damage taken becomes meaningful for a modestly invested life pool:

$$\text{Smith below 90\%: } \frac{100,000}{100,000 + 10H} = 0.90 \quad \Rightarrow \quad H \approx 1,111$$

Shaman remains above 90% mitigation through approximately $H \approx 2,600$ (interpolated between the 92.8% at $H=2,000$ and 88.3% at $H=4,000$ table values). Shaman maintains the “highly efficient” mitigation regime for more than double the hit size that Smith can sustain before the same threshold is crossed.

5 The 2,000 HP Benchmark and Survival Thresholds

Minimum Survival at the Endgame Hit Ceiling

The largest single physical hits in Path of Exile 2 endgame content approach 10,000 raw damage. Against this ceiling, Shaman’s full defensive stack produces 1,923 damage taken. A character with a 2,000 effective HP pool survives with 77 HP remaining (3.9% buffer). This is not a marginal achievement — 2,000 effective HP is a modest endgame life pool attainable without any dedicated life-stacking investment.

Table 5: Survival Analysis — 10,000 Raw Physical Hit (Endgame Ceiling)

Build	Damage Taken	Min HP to Survive	Survives at 2,000 HP?	HP Remaining
Baseline (25k arm.)	8,000	8,001	No	-7,923
Smith of Kitava (100k)	5,000	5,001	No	-4,923
Shaman (75% conv.)	1,923	1,923	Yes	77
Shaman (min-maxed)	1,388	1,389	Yes (comfortable)	1,611

Min-maxed Shaman assumes 35,000 armour and 90% max resistance. “HP Remaining” is effective HP pool minus damage taken; negative means the character dies on the hit. Smith would need approximately 5,000 effective HP to guarantee survival of the same hit, a requirement demanding heavy life, ES, or Mind over Matter investment on top of the already heavy armour commitment.

The Mind over Matter Compound

Mind over Matter (MOM) redirects a portion of incoming damage to mana before it reaches life, creating an effective HP buffer from the mana pool. This interacts favourably with the Shaman’s low final damage output:

With 40% of damage redirected to mana, a 1,923 total damage hit splits as:

$$\text{Damage to life: } 1,923 \times 0.60 = 1,154 \quad \text{Damage to mana: } 1,923 \times 0.40 = 769$$

A Shaman with 1,500 life and 800 mana survives the endgame ceiling hit: life absorbs 1,154 (survives at 346 HP), mana absorbs 769 (depleted but absorbed). The *required mana pool to cover the MOM redirect* is only 769 — a small number that does not demand dedicated investment.

Smith attempting MOM against a 5,000-damage hit:

$$\text{Damage to life: } 5,000 \times 0.60 = 3,000 \quad \text{Damage to mana: } 5,000 \times 0.40 = 2,000$$

Smith now requires both 3,001 life *and* 2,001 mana simultaneously — consuming gear affixes and passive points in two separate stats while still wearing the heaviest armour investment possible. MOM is a multiplier on Shaman's already-low damage intake; it is a band-aid on Smith's fundamental survivability problem.

6 Armour Efficiency Amplification

What “Quadruples the Armour Pool” Means Precisely

The conversion reduces the effective physical hit to 25% of the raw value. Because the armour formula evaluates against hit size rather than raw damage, this is equivalent to multiplying armour rating by $4\times$ for physical reduction purposes:

$$\frac{A}{A + 10 \times \frac{H}{4}} = \frac{4A}{4A + 10H} \iff \text{effective armour} = 4 \times A$$

Explicitly at $H=10,000$:

$$\text{Shaman: } \frac{25,000}{25,000 + 2,500} \times \frac{1}{0.25} \approx \frac{100,000}{200,000} = 50\% = \text{Smith: } \frac{100,000}{200,000}$$

Both achieve 50% physical DR on their respective physical components. Shaman uses 25,000 armour; Smith uses 100,000 armour. The conversion did the work of 75,000 additional armour rating.

The Armour Equivalence for Total Mitigation

Matching the Shaman’s *total* effective mitigation (80.8%) through armour alone, with no conversion and 75% resistance on any elemental incidental damage:

$$\frac{A}{A + 100,000} = 0.808 \implies A = \frac{0.808 \times 100,000}{1 - 0.808} \approx 421,000$$

421,000 armour is not achievable in Path of Exile 2. The mitigation ceiling of Shaman’s 25,000 armour plus 75% conversion cannot be replicated through armour investment at any practical scale.

7 Full Defensive Layer Synergy

The hit-size tables in Section 3 use conservative assumptions: they include conversion, armour, armour-to-elemental, and resistance, but omit the multiplicative “less elemental damage taken” ascendancy notable and block. Both of these layers compress the final taken number further.

Table 6: **Shaman Defensive Layer Summary — All Layers**

Layer	Function	Shaman Access
Phys-to-Ele Conversion	Reduces effective physical hit size; feeds elemental layers	75% via Cloak + ascendancy + tree notable + anoint
Armour (physical DR)	Mitigates the 25% residual physical hit	50% DR at 25k armour vs 2,500 effective hit
Armour-to-Elemental	Extends armour formula to each converted ele hit independently	100%+ via ascendancy nodes
Max Elemental Resistance	Caps each element type at 85–90%	Primal Aegis ascendancy path
Less Elemental Taken	Multiplicative reduction after resistance — stacks with all above	Dedicated ascendancy notable
Block	Full hit avoidance before all mitigation layers	At or near maximum cap via passive placement
Hybrid eHP (Life / Mana / ES)	Expands effective health pool beyond raw life	MOM + EB accessible on Shaman passive pathing
Combined	Elemental component near-zero; physical DR 4× peers	All layers within normal progression budget

MOM = Mind over Matter; EB = Eldritch Battery. The “less elemental taken” multiplier is not included in the Section 3 tables; it provides an additional 15–25% multiplicative reduction on the fire component (635 damage at $H=10,000$), pushing total mitigation from 80.8% to an estimated 82–85%. Min-maxed configuration with this layer would approach 87–89% total mitigation at the endgame ceiling.

The Elemental Collapse in Detail

At H=10,000, the converted elemental portion (7,500 raw) is handled across three sequential reduction layers:

1. **Armour-to-Elemental:** The 6,000 fire hit faces 29.4% DR from 25,000 armour, leaving 4,235. Each 500-damage cold/lightning/random hit faces 83.3% DR, leaving only 83 damage each.
2. **Max Resistance (85%):** The 4,235 post-armour fire becomes 635. The 83 per-element remnants become 12 each (36 total for three components).
3. **Less Elemental Taken (est. 20% multiplier):** Compresses the 671 combined elemental to approximately 537.

Against the 25% residual physical (1,250 taken), the elemental component is now less than half of the physical component. The “collapse to near zero” is not hyperbole — 7,500 of converted elemental damage becomes roughly 537 final damage through three sequential reduction layers.

8 Passive Point Economy

Shaman's mitigation superiority is delivered at nearly zero passive budget cost, creating a compound advantage:

1. **Freed points target damage.** The 8–15 passive points Smith commits to armour% clusters become damage nodes, jewel sockets, or life/mana scaling on Shaman. This is not a trade-off — it is a unilateral gain, because the freed points come at no defensive cost (Shaman is already more defensive than Smith).
2. **Gear affixes reallocate.** Armour% suffixes Smith requires become resistance, life, or damage affixes on Shaman. Conversion through ascendancy and a chest item does the work that multiple gear suffixes would otherwise need to cover.
3. **No defensive floor sacrifice.** Because Shaman's mitigation exceeds Smith's at every hit size above roughly $H=1,000$ (the endgame-relevant range), there is no scenario where redirecting passive points to damage creates a defensive shortfall relative to the competing archetype.

Shaman can simultaneously out-defend a committed armour archetype and out-damage (or out-sustain) it through the budget reclaimed from the armour tree.

9 Progressive Investment Breakpoints

Table 7: Shaman Defensive Benchmarks by Progression Stage — H=10,000 Ceiling

Stage	Conv.	Armour	A-to-E	Max Res	Taken	Mitigation
Early Maps	50%	15,000	50%	75%	4,934	50.7%
Mid Red Maps	60%	20,000	75%	80%	3,627	63.7%
Late Endgame	75%	25,000	100%	85%	1,923	80.8%
Min-Maxed	75%	35,000	100%	90%	1,388	86.1%

Early Maps assumes Cloak of Flame equipped but partial ascendancy investment; no passive tree notable or anointment yet. Min-maxed does not include the “less elemental taken” multiplicative layer, which would push effective mitigation to approximately 87–89%. Even at Early Maps, Cloak of Flame alone (50% conversion) exceeds the Baseline build’s mitigation at all hit sizes above $H=500$ and matches or beats Smith of Kitava below $H=2,000$.

The breakpoint table illustrates a key property of this configuration: the Shaman is not a late-game gimmick. Even with Cloak of Flame alone and no other conversion source (Early Maps), the build takes 4,934 damage from a 10,000 hit — equivalent to a Baseline build with roughly 40,000 armour, without having invested a single passive point in armour scaling.

10 Infernalist: Chaos Inoculation and the 20% Hard Immunity

Class Overview

The Infernalist is an ascendancy of the Witch base class and occupies a unique categorical position in the conversion hierarchy. Like Shaman, it accesses physical-to-other-damage-taken conversion through an ascendancy notable — but the target element is chaos rather than elemental, and the interaction with the Keystone Passive Skill **Chaos Inoculation (CI)** transforms what would otherwise be a near-zero elemental reduction into a categorical hard immunity. This distinction, examined precisely, separates Infernalist from every other defensive archetype discussed in this document.

The Infernalist's ascendancy notable *Altered Flesh* converts 20% of all physical damage taken to chaos damage. Combined with Cloak of Flame (50% physical to fire), the baseline conversion stack reaches 70%, leaving 30% residual physical. With CI active, the 20% chaos portion ceases to interact with armour or resistance — it simply does not happen. The build is immune to it.

Conversion Stack: Two Configurations

Table 8: **Infernalist Conversion Stack — Baseline vs. Full Optional Investment**

Source	Type	Element	Baseline	Full (Optional)
Cloak of Flame	Unique Chest	Fire	50%	50%
Molten Being	Passive Tree Notable	Fire	—	5%
Altered Flesh	Ascendancy Notable	Chaos	20%	20%
Catalysis	Anointment	Random Element	—	5%
Total Converted			70%	80%
Residual Physical			30%	20%

Molten Being is located on the northwest side of the passive tree and requires non-trivial passive investment to reach. Catalysis is an anointment consuming the amulet slot. The Full configuration commits both resources, raising fire conversion to 55% and total conversion to 80%. The chaos component (20%) is reduced to 0 under Chaos Inoculation regardless of configuration — it does not enter any further calculation. Fire sub-total in Full = 50 + 5 = 55%; chaos = 20% (immune); random = 5%; residual physical = 20%.

The Categorical Distinction: Hard Immunity vs. Near-Zero Reduction

All previous builds in this document reduce elemental contributions to near-zero through sequential layers (armour-to-elemental, resistance, less-elemental-taken). “Near-zero” is not zero. At $H=10,000$, Shaman’s cold, lightning, and random elemental components each resolve to approximately 12 damage after all layers — small, but nonzero, and the total elemental contribution still represents 671 damage before the “less elemental taken” multiplier.

The Infernalist with CI operates differently on the chaos component. The 20% of the hit that converts to chaos does not pass through any mitigation layer. It is avoided in its entirety:

$$\text{Chaos damage taken} = 0.20 \times H \times (1 - \text{CI immunity}) = 0.20 \times H \times 0 = 0$$

At $H=10,000$, CI avoids **exactly 2,000 damage** — hard, not as a limit approached asymptotically. This is also the most reliable defensive statement possible: chaos immunity does not degrade with hit size, does not depend on armour values or resistance caps, and cannot be bypassed by resistance penetration or reduction. The 20% disappears categorically regardless of what the hit does.

This also addresses chaos damage from other sources entirely. In current Path of Exile 2 endgame content, chaos damage is among the most lethal because it bypasses most conventional mitigation. CI converts this from the build’s most dangerous threat into its most completely nullified one.

Hit-Size Comparison: Infernalist vs. Shaman

The following table uses the same build parameters as prior sections (25,000 armour, 100% armour-to-elemental, 85% max resistance), with CI accounting for the full 20% chaos exemption.

Why the Baseline Underperforms Shaman Despite CI

At first glance, negating 20% of a 10,000 hit (2,000 damage hard avoided) should place Infernalist ahead of Shaman. The number shows the opposite: Infernalist Baseline takes 2,136 versus Shaman’s 1,922. The reason is the physical residual:

Table 9: **Damage Taken and Mitigation — Shaman vs. Infernalist Configurations**

Hit	Shaman Tkn	Sham%	Inf. Base Tkn	IB%	Inf. Full Tkn	IF%
100	<1	99.5%	<1	99.5%	<1	99.7%
500	11	97.8%	12	97.6%	8	98.4%
1,000	41	95.9%	45	95.5%	30	97.0%
2,000	143	92.8%	159	92.1%	106	94.7%
4,000	469	88.3%	523	86.9%	351	91.2%
8,000	1,387	82.7%	1,545	80.7%	1,054	86.8%
10,000	1,923	80.8%	2,136	78.6%	1,469	85.3%

“Inf. Base” = Infernalist baseline: 50% fire, 20% chaos (CI = 0), 30% physical. “Inf. Full” = Infernalist full: 55% fire, 20% chaos (CI = 0), 5% random, 20% physical. Shaman column reproduced from Section 3 for reference. All configurations use 25,000 armour, 100% armour-to-elemental, 85% max elemental resistance. Note that the Infernalist Baseline takes more damage than Shaman at all hit sizes despite CI, because the larger physical residual (30% vs 25%) more than offsets the chaos exemption at this armour level. The Full configuration resolves this and outperforms Shaman at every hit size.

- Shaman: 25% residual physical (2,500 effective hit) → armour DR = 50% → 1,250 taken
- Infernalist Base: 30% residual physical (3,000 effective hit) → armour DR = 45.5% → 1,636 taken

The 386-unit increase in physical damage taken (1,636 vs 1,250) exceeds the 500-unit reduction from the CI fire-contribution shift. CI is extremely efficient at avoiding the chaos component — but the baseline conversion stack leaves enough physical residual that the armour formula’s larger denominator costs more than CI saves, at this armour value.

This resolves cleanly when the Full optional stack is active. Reducing physical residual to 20% (2,000 effective hit) gives 55.6% armour DR and 889 damage taken — well below Shaman’s physical component — producing the 1,469 total that outperforms Shaman.

Accelerating Returns on Conversion: The Scaling Efficiency Argument

Each additional 5 percentage points of physical-to-other conversion is worth *more* than the previous 5 percentage points. This is a direct consequence of the fact that the

gain is calculated against a shrinking denominator (the remaining physical residual):

$$\text{Gain of next 5pp} = \frac{\Delta \text{Residual}}{\text{Residual Before}} = \frac{5\%}{\text{Current Residual}}$$

Table 10: **Accelerating Returns: Relative Value of Each 5 pp Conversion Increment**

Conversion Step	Residual Before	Residual After	Physical I
60% → 65%	40%	35%	12.5% les
65% → 70%	35%	30%	14.3% les
70% → 75%	30%	25%	16.7% (
75% → 80%	25%	20%	20.0% (Inferna
80% → 85%	20%	15%	25.0% les
85% → 90%	15%	10%	33.3% les
90% → 95%	10%	5%	50.0% les
Key Insight	Each 5 pp step is more valuable than the last — gains accelerate as c		

“Physical Reduction Gain” is the percentage reduction in residual physical damage per hit relative to the previous step. This is not a marginal effect — moving from 75% to 80% conversion (Infernalist Full’s contribution) produces 20% less physical damage taken per hit, compared to 16.7% for the Shaman’s 70% → 75% step. The same logic applies in both directions: viewed in reverse, failing to acquire the next 5 pp of conversion costs increasingly more as you approach high total conversion. The Infernalist Full’s 75% → 80% step is categorically more valuable than the Shaman’s equivalent preceding step.

This table validates the user’s intuition precisely. The Shaman’s 70% → 75% step was identified as a “monumental leap” — 16.7% less physical damage per hit. The Infernalist Full’s 75% → 80% step is worth 20.0%, 20% more gain than the Shaman’s step, confirming that it represents a strictly greater efficiency increment, not merely an equivalent one.

Opportunity Cost: Molten Being vs. Energy Shield

The Full configuration requires Molten Being (passive tree notable, non-trivial pathing cost) and Catalysis (anointment slot, consuming the amulet annoint). The user correctly frames this as an opportunity cost decision rather than a clear win:

- **Passive points for Molten Being** could alternatively deepen the Energy Shield pool, increasing the effective HP ceiling for CI. A larger ES pool survives larger hits before the mitigation percentage even applies.

- **The anointment slot** is consumed by Catalysis; other anointments providing ES, resistance, or damage may provide greater net value depending on the build’s current gaps.
- **However:** The 20% reduction in physical residual from Infernalist Full vs. baseline (shown in the hit-size comparison table above) reduces the survival HP requirement from 2,137 to 1,469 — a 668 HP difference at H=10,000. Whether 668 ES from the anointment alternative exceeds this survival margin depends entirely on build-specific ES totals.

At moderate ES pools (<4,000), the Full configuration likely provides a superior marginal survival guarantee per passive point than additional ES would. At very large ES pools (>6,000), the baseline configuration’s survival floor is already comfortable, and the Molten Being path may be less efficient than ES amplification. The decision is build-stage dependent.

Infernalist Positioning

Table 11: **Infernalist Defensive Profile vs. Shaman at H=10,000**

Configuration	Conv.	Damage Taken	Mitigation	Min HP
Shaman (standard)	75%	1,923	80.8%	1,923
Infernalist Baseline (CI)	70% + CI	2,136	78.6%	2,136
Infernalist Full (CI)	80% + CI	1,469	85.3%	1,469
Shaman Min-Maxed	75%	1,388	86.1%	1,388
Infernalist Full Min-Maxed	80% + CI	est. <1,100	est. >89%	est. <1,100

All entries at 25,000 armour, 100% armour-to-elemental, 85% max resistance unless noted. Min-Maxed Shaman uses 35,000 armour and 90% max resistance. Infernalist Full Min-Maxed is estimated, incorporating higher armour (35,000), 90% max resistance, and the “less elemental taken” multiplicative layer; CI immunity on the chaos component is maintained regardless of other parameters. The Infernalist Full at standard investment surpasses the Shaman at standard investment and approaches the Shaman Min-Maxed ceiling — while being supported by an ES pool (CI) rather than a conventional life pool, and providing categorical chaos immunity as an additional defensive property.

The Infernalist represents a genuine alternative defensive powerhouse rather than a lateral trade. Its distinguishing properties relative to Shaman are: (1) a hard chaos immunity rather than a near-zero residual on the converted component, (2) an ES-only effective HP pool under CI that interacts favorably with Energy Shield mechanics, and (3) access to spell-casting, minion, and select martial weapon playstyles, with

the ascendancy favouring fire and spell-caster builds most heavily. The baseline configuration is slightly below Shaman in raw physical mitigation but offers chaos immunity as compensation; the Full configuration exceeds Shaman in raw physical mitigation while retaining that immunity, at the cost of committed passive investment.

11 Invoker: Avoidance, Evasion-as-Armour, and the Conditional eHP Ceiling

Class Overview and Defensive Paradigm

The Invoker is an ascendancy of the Monk base class and represents a categorically different defensive architecture from the conversion builds documented in preceding sections. Where Shaman and Infernalist achieve passive per-hit mitigation by suppressing the effective hit size through physical-to-elemental conversion, the Invoker achieves defensive efficiency through *avoidance* — preventing hits from connecting at all — backed by a novel ascendancy mechanic that converts evasion rating into armour for the hits that do land.

This distinction carries a practical cost that gives the build its informal label: **the coward's build**. Fully realising the Invoker's defensive ceiling requires a pre-combat ritual — the *Meditate* skill, which overflows the Energy Shield pool to 1.5× its base value. A player who cannot or does not meditate before engaging a threatening encounter forfeits the primary survival buffer against large hits. This playstyle constraint is non-negotiable at high investment levels: the build is designed around strategic engagement, not facetanking.

In exchange for this constraint, the Invoker achieves gear and passive point efficiency that rivals Shaman's, through a completely different mechanism: near-zero need for life affixes, chaos resistance affixes, or spirit affixes on gear, freeing the entire affix budget for evasion and Energy Shield. The resulting eHP pool, amplified by *Meditate*, is conditionally larger than what most other archetypes can reach at comparable investment.

The Three Mandatory Ascendancy Nodes

The Invoker has four ascendancy choice slots. Three are near-mandatory for the defensive architecture described in this section; the fourth is available for damage or situational utility.

Meditate: The Conditional eHP Ceiling

Meditate, granted by *Faith is a Choice*, allows the Invoker to overflow their Energy Shield to 1.5× its maximum value. This overflow cannot be achieved through regener-

Table 12: **Invoker Ascendancy Nodes — Defensive Core**

Node	Effect	Priority
Faith is a Choice	Grants the Meditate skill. Doubles ES regeneration rate. ES can overflow to $1.5\times$ maximum via Meditate between encounters.	Mandatory
Lead me through grace...	Gain spirit for each X points of combined evasion and Energy Shield on equipment. Cannot benefit from explicit spirit modifiers on equipment.	Mandatory (gate)
...and protect me from harm	35% less evasion rating (final multiplier, not overcome by increased evasion). Armour DR is calculated from combined armour <i>and</i> evasion pool.	Mandatory (key node)
Fourth Choice	Damage, utility, or situational defensive node. Outside mandatory defensive core.	Situational

“Lead me through grace...” is the gating node that unlocks “...and protect me from harm.” Both must be taken to access the armour-evasion unification, making them functionally a paired investment. The “35% less” modifier on evasion is a final multiplier in the evasion rating formula and cannot be negated by increased evasion rating modifiers on passive tree or gear.

ation or flask mechanics alone — it requires actively channelling Meditate outside of combat.

The practical impact is that the Invoker operates at two distinct eHP levels:

- **Base (in combat):** The natural ES pool. This is the floor and is the value that matters if the player engages without preparation.
- **Meditate (pre-combat):** $1.5\times$ the base ES pool. This is the ceiling and is what the build's survival thresholds are calibrated against.

Against a hit that would take 4,850 damage when it lands (derived from the armour-evasion DR layer), a character needs a base ES of 4,850 to survive without Meditate, but only 3,234 ES with Meditate active ($4,850 / 1.5 = 3,233$). Meditate effectively reduces the required raw ES investment by 33% for any given survival threshold — a significant return on the single ascendancy point.

The build-design implication is that the ES pool should be calibrated to the Meditate value, not the base value. Investment decisions should ask: “at $1.5\times$ my projected ES, do I survive the relevant hit ceiling?” rather than targeting the base pool.

Chaos Inoculation and Gear Architecture

Like the Infernalist, the Invoker is the natural beneficiary of the **Chaos Inoculation** keystone passive:

Maximum life = 1 Immune to chaos damage

For the Invoker, CI delivers three compounding advantages:

1. **Chaos damage immunity.** As with the Infernalist's Altered Flesh interaction, chaos damage is the most lethal unmitigated source in current endgame content. CI converts the Invoker's greatest potential vulnerability into its most completely nullified threat.
2. **No chaos resistance required.** Capping chaos resistance conventionally consumes multiple gear suffixes and rune slots. CI frees every one of these for other affixes — at zero ongoing opportunity cost.
3. **No life affixes required.** With life set to 1 by CI, life investment is irrelevant. Every gear prefix that would otherwise go to life or life% becomes available for flat Energy Shield or increased Energy Shield.

The net effect on gear architecture: every affix slot on every piece of equipment is available for flat evasion, local evasion rating%, flat Energy Shield, and increased Energy Shield. No life, no chaos resistance, no spirit (blocked by the ascendancy). This is one of the most focused and unconstrained affix budgets of any archetype in the game.

“Lead me through grace...” — Spirit via Evasion and ES

This ascendancy node delivers spirit — the resource that enables active skills, auras, and minions — based on evasion and Energy Shield values on equipment, rather than from explicit spirit affixes. Simultaneously, it *prevents* the character from benefiting from explicit spirit modifiers on equipment.

The apparent detriment (no explicit spirit gear) is in practice a benefit. Spirit is a highly sought-after and contested affix on gear; other archetypes compete for it and must accept the opportunity cost of taking it over other affixes. The Invoker is *incentivised not to pursue it*, because the ascendancy generates spirit from what the Invoker already wants: high evasion and high Energy Shield values on gear. This creates a secondary reward loop:

- Pursuing evasion+ES gear (for defensive value) also passively generates the spirit budget
- That spirit budget enables *Galeforce* and other auras without consuming any affix slots
- Evasion+ES gear is generally easier to craft and less contested in trade than spirit-explicit gear
- The freed spirit affix slot becomes another evasion or ES affix

The Invoker resolves two resource pressures simultaneously (spirit and defensive stats) with a single gear pursuit. Other archetypes must solve them separately.

“...and protect me from harm” — The Armour-Evasion Unification

This is the key defensive node and the primary reason the Invoker belongs in a document about defensive efficiency. Its two effects:

1. **35% less evasion rating** (final multiplier on the evasion formula)

2. Armour DR is calculated from combined armour and evasion pool

The first effect is the cost; the second is the payoff. Critically, the “less” modifier reduces evasion rating (not evasion chance directly), and the armour calculation uses the same post-less evasion value:

$$\text{Effective Evasion} = \text{Base Evasion} \times 0.65 \quad (\text{for both dodge chance and armour})$$

$$\text{Effective Armour} = \text{Base Armour} + \text{Base Evasion} \times 0.65$$

At 50,000 base evasion with 5,000 base armour from shields and gear:

$$\text{Effective Evasion} = 32,500 \quad \text{Effective Armour} = 5,000 + 32,500 = 37,500$$

The character loses 35% of their evasion rating (an avoidance reduction), but gains 32,500 effective armour from the same pool. The net exchange is: *slightly fewer hits avoided, but hits that do land face meaningful DR from free armour*. At high base evasion values, the armour gain dominates the evasion loss in terms of survivability value.

DR from Evasion-Derived Armour: Hit-Size Table

The following table uses the armour DR formula ($A/(A+10H)$) applied to the effective armour pool derived from evasion, at varying base evasion investment levels. Base armour of 5,000 is assumed from shields and incidental gear.

Optimal Evasion Targets per ES Pool

The conventional armour optimisation heuristic states that effective armour should fall in the range of $12\times$ – $20\times$ the effective HP pool to provide efficient DR against the range of hits a character encounters. For the Invoker, “effective armour” is derived from evasion, allowing a direct calculation of the evasion target per ES pool size.

The design target for an Invoker can therefore be expressed as a simple paired constraint: acquire evasion equal to $12\times$ your base ES (at minimum) so that each point of evasion investment simultaneously improves dodge chance AND produces armour in the efficient DR range. Any evasion beyond the $20\times$ ceiling still improves avoidance but no longer improves DR meaningfully — the investment shifts category

Table 13: **Armour DR from Evasion-as-Armour by Hit Size (5,000 base armour + evasion $\times$ 0.65)**

Base Evs	Eff. Arm	H=100	H=500	H=1k	H=2k	H=4k	H=8k	H=10k
20,000	18,000	94.7%	78.3%	64.3%	47.4%	31.0%	18.4%	15.3%
30,000	24,500	96.1%	83.1%	71.0%	55.1%	38.0%	23.4%	19.7%
40,000	31,000	96.9%	86.1%	75.6%	60.8%	43.7%	27.9%	23.7%
50,000	37,500	97.4%	88.2%	78.9%	65.2%	48.4%	31.9%	27.3%
70,000	50,500	98.1%	91.0%	83.5%	71.6%	55.8%	38.7%	33.6%
100,000	70,000	98.6%	93.3%	87.5%	77.8%	63.6%	46.7%	41.2%

Cell shading uses the same tier scale as Section 3: dark green $\geq 95\%$, light green 90–94%, yellow 80–89%, orange 70–79%, red $< 70\%$. Unlike Shaman and Infernalist, whose mitigation columns hold in the yellow-green band at large hits, the Invoker’s DR column falls into red above $H=4,000$ at all practical evasion values. This is by design: the Invoker’s defence against large hits is avoidance (not being hit at all), not passive DR. The DR layer handles the inevitable hits that slip through evasion and block.

Table 14: **Optimal Base Evasion Targets by ES Pool (12 $\times$ –20 $\times$ DR Optimality Range)**

ES Pool	Meditate ES	Arm. Low (12 $\times$)	Arm. High (20 $\times$)	Base Evs (Low)	Base Evs (High)
2,000	3,000	24,000	40,000	29,231	47,692
3,000	4,500	36,000	60,000	44,847	74,538
4,000	6,000	48,000	80,000	60,463	100,384
5,000	7,500	60,000	100,000	76,079	126,230
6,000	9,000	72,000	120,000	91,695	152,076

“Base Evs” is the pre-35%-less evasion rating required so that evasion $\times$ 0.65 (plus 5,000 base armour) equals the armour target. The optimality range applies to the base ES pool, not the Meditate value; Meditate is shown separately as the survivable ceiling. A 3,000 ES Invoker needs approximately 48,000–85,000 base evasion to keep armour-from-evasion in the efficient DR range. Above these evasion values, additional evasion contributes primarily to avoidance chance rather than marginal DR improvement. Meditate ES is shown to illustrate the conditional survival ceiling: a 3,000 ES Invoker with Meditate has 4,500 effective eHP at encounter entry.

rather than becoming wasted.

Survival Thresholds: When a Hit Lands

When a hit penetrates both evasion and block, the Invoker faces the raw damage reduced only by evasion-derived armour. At 50,000 base evasion (37,500 effective armour), the minimum ES required to survive each endgame hit tier is:

Table 15: **ES Survival Requirements When a Hit Lands — 50,000 Base Evasion (37,500 Eff. Armour)**

Hit Size	DR	Damage Taken	Min ES (Base)	Min ES + Meditate	Meditate Saves
1,000	78.9%	211	211	141	70
2,000	65.2%	696	696	464	232
4,000	48.4%	2,065	2,065	1,377	688
8,000	31.9%	5,447	5,447	3,631	1,816
10,000	27.3%	7,273	7,273	4,849	2,424

“Min ES + Meditate” is the minimum base ES required when Meditate overflow (1.5×) is active at encounter entry: damage taken /1.5. “Meditate Saves” is the reduction in required base ES attributable solely to the overflow. At H=10,000, Meditate reduces the required base ES from 7,273 to 4,849 — saving 2,424 ES of investment. Without Meditate, surviving the endgame hit ceiling at this evasion level would require a 7,273 ES pool, which is unrealistic for most builds. With Meditate, 5,000 ES (7,500 meditate) provides a buffer. This is the mechanical justification for the “coward’s build” label: Meditate is not optional for surviving the hardest content.

The table makes explicit what the playstyle label implies: the Invoker’s survival against ceiling-tier hits is *contingent* on Meditate. A 5,000 ES Invoker who enters a boss room without meditating faces a 7,273 damage hit on a 5,000 ES pool and dies. The same character who meditates has 7,500 effective eHP and survives by 227 points. The margin is not large, which is why the 12×–20× evasion optimality range and ongoing ES investment remain important — they push the meditate ceiling higher and provide more room for error.

Combined Avoidance: Evasion and Block

The Invoker’s primary defence against all hits — including the largest — is not mitigating them but preventing them from connecting. With a spear and buckler setup providing near-cap block (50–65%) layered over evasion chance (50–65%), the combined hit avoidance is:

$$P(\text{hit avoidance}) = 1 - (1 - e)(1 - b)$$

Table 16: **Combined Hit Avoidance (%) — Evasion Chance × Block Chance**

Evs \ Block	30%	40%	50%	55%	60%	65%
40%	58.0%	64.0%	70.0%	73.0%	76.0%	79.0%
50%	65.0%	70.0%	75.0%	77.5%	80.0%	82.5%
60%	72.0%	76.0%	80.0%	82.0%	84.0%	86.0%
65%	75.5%	79.0%	82.5%	84.2%	86.0%	87.8%
70%	79.0%	82.0%	85.0%	86.5%	88.0%	89.5%
75%	82.5%	85.0%	87.5%	88.8%	90.0%	91.3%

At the achievable baseline of 50% evasion and 50% block, 75% of all hits are avoided entirely before any mitigation layer applies. At 65% evasion and 65% block (requiring significant investment), avoidance reaches 87.8%. Pushing both values to 75% achieves 90%+ avoidance. The 98–99% avoidance threshold implied by optimal values requires extreme investment in both stats simultaneously and may not be achievable within the passive and gear budget while maintaining the ES pool targets described above. The practical target for a well-invested Invoker is 82–88% combined avoidance, meaning 1 in 6–5 hits connects. Those connecting hits are handled by the evasion-derived armour and Meditate ES pool.

Per-Encounter Damage Expectation vs. Shaman

The Invoker and Shaman arrive at comparable per-encounter defensive outcomes through opposite mechanics. Consider 20 hits at H=1,000 in a typical encounter:

- **Shaman:** Every hit connects. Each hit takes 40.6 damage. Total encounter damage: $20 \times 40.6 = 812$.
- **Invoker (50% evs, 50% block):** 75% of hits are avoided. Expected hits connecting: $20 \times 0.25 = 5$. Each landing hit takes 211 damage. Total encounter damage: $5 \times 211 = 1,055$.

On expected value, the Invoker takes slightly *more* total damage per encounter at H=1,000 (1,055 vs 812). However, the Invoker’s defence profile produces dramatically different variance: most encounters end with zero damage (all hits avoided), but the build is exposed to burst clustering where multiple consecutive hits land before evasion resets. Shaman’s profile is flat — it takes the same amount of damage regardless of hit sequencing.

This is the core trade-off: Shaman is statistically smooth; Invoker is statistically bursty.

The Invoker mitigates burstiness through Meditate’s pre-combat eHP overhead and block’s independent avoidance layer, but the risk profile is fundamentally different. At H=10,000 (the ceiling where survival matters most), the Invoker with full Meditate is not guaranteed to survive a landed hit unless the ES pool and evasion-armour are correctly calibrated.

Supporting Systems and Future Exploration

Several additional mechanics compound the Invoker’s defensive and offensive efficiency but fall outside the quantitative scope of this document:

- **Galeforce:** An aura that grants stacking evasion rating increases, funded by the spirit pool generated passively through “Lead me through grace...” Because the spirit pool scales with gear evasion+ES values (which the Invoker is already maximising), Galeforce is effectively a zero-cost aura. Its stacks push base evasion rating higher during combat, improving avoidance and increasing the effective armour value in the armour-evasion formula.
- **Deflection:** A mechanic that applies a front-end multiplicative reduction to hit damage before armour is calculated. If accessible to the Invoker, deflection would reduce the effective hit size entering the armour formula, improving DR at every hit tier — analogous to how physical-to-elemental conversion reduces the effective hit size for Shaman. Exploring the interplay between deflection and evasion-derived armour is a priority for future analysis.
- **Hollow Palm:** A keystone or technique available to unarmed Monk builds that scales defences (and damage) from having no weapon equipped. Whether a Hollow Palm Invoker can achieve comparable or superior evasion-armour targets through a different gearing axis is worth investigating, particularly given the freed gear slots.
- **Spear and Buckler Synergy:** The reference configuration (spear + buckler) provides the near-50% block baseline. Combined with the evasion values described above, this produces the 82–88% avoidance band. Exploring whether shield-focused passive nodes can push block toward 60–65% while preserving evasion investment is likely the highest-leverage path to improving the Invoker’s ceiling, as 65% evasion + 65% block (87.8% avoidance) represents a substantially different risk profile than the 50%+50% baseline.

Invoker Positioning

Table 17: Invoker vs. Shaman: Defensive Profile Comparison

Property	Shaman	Invoker
Primary defence	Passive hit mitigation (conversion + layers)	Avoidance (evasion +
DR on a landed hit (H=10k)	80.8%	27.3% (50k evs) – 41.2
eHP source	Life (+ MOM/EB optional)	Energy Shield (CI mar
eHP conditional boost	Mind over Matter (passive)	Meditate (active pre-c
Chaos damage	Near-zero via resistance	Zero (CI immunity)
Gear affix budget	Frees armour affixes	Frees life, chaos res, sp
Passive point spend	Damage nodes (conversion from ascendancy)	Evasion+ES nodes (de
Playstyle constraint	None	Meditate before challe
Ceiling comparison	Survives H=10k on 1,923 HP (passive)	Survives H=10k with a
Verdict	Passive survivability, lower eHP floor	Conditional higher

The two archetypes are not directly comparable by a single metric. Shaman guarantees survival of the H=10,000 ceiling on 1,923 HP passively. Invoker guarantees avoidance of ~75–88% of all hits, with survival of landed hits conditional on Meditate. The Invoker’s strength is encounter-level expected damage (most hits miss) and a larger achievable eHP ceiling via Meditate overflow; Shaman’s strength is unconditional per-hit survivability and passive simplicity. Players optimising for smoothness of play and reliability under all conditions favour Shaman; players optimising for peak eHP ceiling and total encounter damage intake under optimal play favour Invoker.

12 Witchhunter: Sorcery Ward and the Absorption-Layer Paradigm

Class Overview and Defensive Paradigm

The Witchhunter is an ascendancy of the Mercenary base class and represents the third distinct defensive architecture examined in this analysis. Where Shaman and Infernalist achieve mitigation by suppressing the effective hit size through physical-to-elemental conversion, and where the Invoker achieves efficiency through avoidance, the Witchhunter sidesteps the armour formula's structural weakness—its quadratic degradation against large hits—through a fundamentally different mechanism: an absorption layer that intercepts post-armour damage before it reaches the life pool.

This distinction is categorical. Damage reduction (DR) scales as $A/(A + 10H)$, which collapses toward zero as H grows regardless of armour investment. An absorption pool absorbs a flat quantum of post-DR damage without degrading with hit size. Against very large hits, the Witchhunter's Sorcery Ward (SW) therefore provides more reliable protection per point of investment than additional armour provides. The Witchhunter achieves this without a unique-item dependency, making its defensive model fully accessible in Solo Self-Found, Hardcore, and HC-SSF contexts.

The Two Mandatory Ascendancy Nodes

The Witchhunter's absorption stack requires two of its four ascendancy notables. These are non-negotiable; the remaining two slots are available for offensive or utility investment.

Sorcery Ward Mechanics

Sorcery Ward operates as an absorption layer placed between the armour DR step and the life pool in the damage sequence:

$$\text{SW Pool} = \min(32,000, 0.30 \times (\text{Post-Penalty Armour} + \text{Post-Penalty Evasion}))$$

$$\text{Effective Armour (post-penalty)} = \text{Geared Armour} \times 0.50$$

Table 18: **Witchhunter Ascendancy Nodes — Defensive Core**

Node	Effect	Priority
Obsessive Rituals	50% less to armour and evasion ratings. Grants Sorcery Ward: absorption pool equal to 30% of post-penalty combined armour+evasion, capped at 32,000.	Mandatory
Ceremonial Ablution	Extends SW coverage from elemental hit damage to <i>all</i> hit damage types, including physical and chaos.	Mandatory
Third/Fourth Choice	Offensive scaling, utility, or situational defensive investment.	Situational

Obsessive Rituals is the gating node for the SW mechanic; Ceremonial Ablution converts it from an elemental-only to an all-damage absorption pool. Both must be taken to achieve full-damage-type coverage. The 50% less multiplier on armour and evasion is a final multiplier, not overcome by increased armour nodes; effective armour post-notables is exactly half of geared armour.

Damage Sequence: Armour DR $\rightarrow$ SW absorbs remainder $\rightarrow$ Life receives overflow

SW recharges fully on a 6-second cooldown. Once depleted by a hit, SW is unavailable until the cooldown expires — making the build vulnerable to rapid burst clusters that deliver multiple large hits within the 6-second window.

At 60,000 geared armour: post-penalty effective armour = 30,000. Sorcery Ward = $0.30 \times 30,000 = 9,000$. At $L = 3,000$ HP, total absorption stack = $9,000 + 3,000 = 12,000$.

Reaching the 32,000 SW cap requires a post-penalty armour+evasion pool of approximately 107,000, or approximately 213,000 before the 50% penalty — confirming that 9,000 is a realistic and achievable mid-endgame figure.

Hit-Size Survival Table

The following table applies $DR = A/(A + 10H)$ at the post-penalty armour of 30,000, then deducts the post-DR remainder from $SW = 9,000$, with overflow reaching life at $L = 3,000$.

Table 19: **Witchhunter Survival — 60,000 Geared Armour (30,000 Eff.), SW = 9,000, L = 3,000 HP**

Raw Hit	DR	Post-DR	SW Absorbs	Life Dmg	Survives?
5,000	37.5%	3,125	3,125	0	Yes
8,000	27.3%	5,818	5,818	0	Yes
10,000	23.1%	7,692	7,692	0	Yes
12,000	20.0%	9,600	9,000	600	Yes
14,000	17.6%	11,529	9,000	2,529	Yes
14,485	17.1%	12,007	9,000	3,007	Threshold

$A = 30,000$ (post 50% less), $SW = 9,000$, $L = 3,000$. Survival ceiling is derived from setting life damage $\leq L$: $10H^2/(30,000 + 10H) - 9,000 = 3,000$, which solves to $H \approx 14,485$. Against the same 10,000 raw hit at equal investment: Smith at 60,000 geared armour and $L = 3,000$ takes approximately 3,509 (exceeding L , structurally fatal at this pool). Shaman at zero armour and $L = 3,000$ takes approximately 3,175 (also exceeding L without additional Energy Shield). Witchhunter takes **zero life damage** from the same hit and survives with full life intact. This is the clearest expression of the absorption paradigm's ceiling advantage.

The survival ceiling of approximately $H = 14,485$ against a raw physical hit substantially exceeds what Shaman or Smith can achieve at comparable investment and life pool. The Witchhunter's SW pool achieves this without any unique-item requirement, passive-point over-investment, or Energy Shield stacking — purely from two ascendancy notables applied to the same armour investment other builds treat as a DR source.

Cross-Damage-Type Coverage

Ceremonial Ablution extends SW coverage to physical, elemental, and chaos hit damage simultaneously. This is structurally superior to the other archetypes' chaos defences:

- **Shaman:** No chaos conversion or absorption layer; chaos damage lands with only resistance mitigation. Chaos is Shaman's exposed damage type.
- **Smith of Kitava:** Dedication to Kitava applies armour DR to chaos hits — effective but limited by the same $A/(A + 10H)$ collapse at large hits.
- **Infernalist/Invoker:** Chaos Inoculation provides complete immunity to chaos damage at the cost of life = 1.
- **Witchhunter:** SW intercepts post-armour chaos damage using the same pool

and the same damage sequence. Against a 10,000 raw chaos hit at $A = 30,000$: DR = 23.1%, post-DR = 7,692, SW absorbs 7,692 — zero life damage. Coverage is equivalent to the physical case shown above.

The Witchhunter is the only archetype in this paper that achieves baseline absorption against physical, elemental, and chaos hit damage from a single two-notable investment, without either requiring a unique item (Shaman: Cloak of Flame) or committing the entire life pool to Energy Shield (CI builds).

Evasion Interaction and the Armour-Evasion Hybrid

Sorcery Ward scales with the *combined* post-penalty armour and evasion pool, making evasion investment doubly efficient for the Witchhunter:

1. Each point of post-penalty evasion contributes 30% of its value to SW, expanding the absorption pool.
2. Evasion provides a hit-avoidance layer that reduces the rate at which the SW pool is consumed per encounter.

An armour/evasion hybrid Witchhunter running a shield achieves the post-penalty armour floor at lower geared investment while simultaneously expanding SW from the evasion contribution. At 50% evasion chance and 50% block chance, 75% of attacks are avoided before the SW pool is engaged — meaning SW is consumed at one-quarter the rate of a pure armour build facing the same encounter. This makes the 6-second recharge cooldown substantially less punishing in practice.

Weapon-Swap Passive Efficiency

Path of Exile 2 allows passive nodes to be allocated to a specific weapon set, becoming active only while that set is equipped. For the Witchhunter, this creates a structural passive-point surplus: nodes on the inactive weapon set cost no investment during the active set's operation.

In practice, a secondary weapon set carries defensive or utility nodes — armour scaling, resistance nodes, SW-contributing clusters — while the primary set runs an entirely offensive allocation. The effective passive budget available to the character exceeds the nominal count, because neither set's nodes compete with the other while they are not simultaneously active.

Two utility nodes are particularly well-suited to the secondary weapon set:

- **Culling Strike:** Kills enemies below a life threshold (typically 10% maximum life). Against enemies that would otherwise require two hits, culling compresses two engagements into one, adding approximately 11% kill efficiency across a full pack. On boss encounters with regenerating life, the execute eliminates the window during which damage must outpace recovery.
- **Explode:** Enemies killed explode dealing a percentage of their maximum life as area physical damage, converting every kill into a secondary clear event. Against dense endgame monster packs, a single killed enemy propagates damage across the remaining pack without an additional skill cast.

Allocated to the secondary weapon set, both nodes cost zero primary-set passive investment; the player retains the full primary-set budget for damage and survivability nodes while gaining substantial clear-speed upside on demand.

Trade-offs

- **Burst vulnerability:** SW recharges on a 6-second cooldown. Rapid consecutive large hits deplete SW on the first strike and leave life exposed until the cooldown completes. Unlike Invoker's Meditate (pre-combat reset) or Shaman's passive DR (always active), SW cannot be refreshed mid-encounter on demand.
- **Damage over time bypass:** Bleed, poison, ignite, and other DoT effects bypass SW and must be managed through flask charges or dedicated mechanics. SW only intercepts hit damage.
- **50% less armour:** The Obsessive Rituals penalty reduces effective armour to half of geared investment. The standard $12\times$ – $20\times$ armour optimality corridor implies a corresponding increase in geared armour targets. However, each additional point of geared armour simultaneously improves DR and expands SW — making armour investment more efficient per passive point for a Witchhunter than for a zero-SW build at the same armour tier.

Witchhunter Positioning

Table 20: **Witchhunter vs. Shaman vs. Invoker: Defensive Profile Comparison**

Property	Shaman	Witchhunter	Invoker
Primary defence	Conversion (DR suppression)	Sorcery Ward (absorption)	Avoidance (evasion +
DR on landed H=10k	80.8% (all layers)	23.1% (armour only)	27.3% (50k evs)
Post-DR absorption	None	9,000 SW (mid-endgame)	None (Meditate pre-co
Survival: H=10k	1,923 HP needed	0 life damage (SW covers)	4,849 ES + Meditate
Chaos damage	Near-zero residual	SW absorbs post-DR	CI immunity (keyston
eHP source	Life (conventional)	Life + SW buffer	Energy Shield (CI)
Unique item required	Yes (Cloak of Flame)	No	No
Playstyle constraint	None	SW cooldown awareness	Meditate pre-combat
Survival ceiling (H)	≈10,000 (standard)	≈14,485 (mid-invest)	Conditional (Meditate
Verdict	Point-efficient, passive	Highest one-shot ceiling	Encounter-level avo

Witchhunter’s survival ceiling is the highest of the three at comparable passive investment, achieved without unique-item dependency. The cost is cooldown vulnerability on burst damage sequences and DoT exposure. Shaman’s profile is passive and consistent; Invoker’s is avoidance-primary with a conditional eHP ceiling via Meditate; Witchhunter’s is absorption-primary with the highest sustainable hit ceiling when SW is charged. Each archetype answers a different threat profile, and each is viable; none is universally superior.

13 Tactician: Deflection, Capped DR, and the Armour-Efficient Paradigm

Class Overview and Build Identity

The Tactician is the second ascendancy of the Mercenary base class and occupies an unusual position in the defensive landscape: its core mechanical identity — supporting totems, crossbows, and minions at range — is intelligible from the passive tree, but the defensive nodes required to access that identity consume the first two ascendancy slots before any offensive expression is possible. For most players, the Tactician is a build that becomes itself only after the campaign ends. This deferred payoff is not a mechanical flaw; it is a design constraint worth naming, because it shapes how one should evaluate the class’s defensive architecture independent of its playstyle.

The two near-mandatory defensive nodes are **Polish that Gear** and **Stay Light, Use Cover**. Together they define an archetype with a hard DR ceiling, a front-loaded damage reduction mechanic, and strong pressure toward ranged engagement. They also create a genuine tension in build optimisation: the DR cap imposed by Stay Light limits how much armour investment matters, the evasion cap it imposes is compounded

by the Glancing Blows keystone, and the primary defensive win condition — deflection — depends on a probabilistic chain that the build can influence but never fully control.

The Two Core Defensive Nodes

Table 21: Tactician Ascendancy Nodes — Defensive Core

Node	Effect	Priority
Polish that Gear	Deflection rating = 20% of armour rating. Ailment threshold = 100% of evasion rating.	Mandatory
Stay Light, Use Cover	Defend with 200% of armour. Enemies have an accuracy penalty based on distance. Maximum chance to evade = 50%. Maximum damage reduction = 50%.	Near-mandatory
Third/Fourth Choice	Build-defining offensive or totem/minion nodes — accessible only after the defensive core is purchased.	Situational

“Polish that Gear” is a prerequisite for “Stay Light, Use Cover.” The “Near-mandatory” label on Stay Light reflects the author’s assessment: the 200% armour defend modifier halves the armour investment required to reach any given DR, and the accuracy-from-distance penalty supports the crossbow/totem identity. A viable alternative skipping Stay Light is analysed in the Build Path Divergence section.

Deflection: The Front-Loaded Multiplier

Polish that Gear grants deflection rating equal to 20% of armour rating. When a hit is deflected, damage is reduced by 40%. The operative question for this analysis — one that the game’s tooltip does not resolve — is whether this reduction applies *before* or *after* the armour formula evaluates.

- **If front-loaded (deflect reduces hit size entering armour):** The armour formula evaluates against $0.6H$ rather than H . The armour formula produces better DR on a smaller effective hit, and the ordering strictly improves outcomes when armour is below the DR cap.
- **If back-loaded (deflect reduces the post-armour remainder):** The armour formula evaluates against the full H . Deflect applies as a multiplier on the damage taken after armour DR.

At the 50% DR cap imposed by Stay Light, the two orderings produce **identical results**:

$$\text{Front-loaded: } 0.6H \times (1 - 0.50) = 0.30H \quad \text{Back-loaded: } H \times (1 - 0.50) \times 0.60 = 0.30H$$

The analysis below treats the DR cap as the planning target, where ordering is irrelevant. Below the cap, front-loading is strictly better: the armour formula operates on a smaller effective hit and yields higher DR at every armour value. A build that consistently reaches the 50% DR cap on relevant hits loses nothing to the ambiguity; one that does not reach the cap benefits more if deflect is front-loaded.

Effective DR on a deflected hit at the cap:

$$\text{Damage taken} = 0.6H \times 0.50 = 0.30H \quad \Rightarrow \quad \text{Effective DR} = 70\%$$

Effective DR on an undeflected hit at the cap:

$$\text{Damage taken} = H \times 0.50 = 0.50H \quad \Rightarrow \quad \text{Effective DR} = 50\%$$

Armour Targets Under 200% Defend

The 200% armour defend modifier means the DR formula evaluates $2A$ against the incoming hit, not A . The standard corridor ($12-20 \times L$) was derived for the unmodified formula; it does not apply here.

$$DR = \min\left(0.50, \frac{2A}{2A + 10H}\right)$$

Cap condition (no deflect): $2A \geq 10H \Rightarrow A \geq 5H$. The armour required to guarantee 50% DR against a raw hit H is exactly $5H$ — half of the standard zero-conversion requirement of $10H$.

Cap condition (with deflect): The formula evaluates against $0.6H$, so $2A \geq 6H \Rightarrow A \geq 3H$.

Table 22: **Revised Armour Corridor — Tactician (200% Defend, 50% DR Cap)**

Scenario	Cap Condition	Vs 2L Kill Threshold	Effective Corridor
Standard (no mod)	$A \geq 10H$	$A = 20L$	$12 \times -20 \times L$
Tactician, no deflect	$A \geq 5H$	$A = 10L$	$5 \times -10 \times L$
Tactician, with deflect	$A \geq 3H$	$A = 6L$	$3 \times -6 \times L$
Planning target (worst-case)	No deflect	$A = 10L$	$\approx 8 \times -10 \times L$

“Kill threshold” assumes 50% DR cap: maximum survivable hit = 2L (half is taken, half is absorbed by pool). Armour required for 50% DR against that hit: $5 \times 2L = 10L$ (no deflect) or $3 \times 2L = 6L$ (with deflect). Because deflection is probabilistic, the prudent planning target is the no-deflect corridor of $8-10 \times L$. This is a structural reduction from the standard $12-20 \times L$ heuristic: the 200% defend modifier provides the equivalent of doubled armour investment, halving the gearing cost to reach the DR cap. Additional armour beyond the cap condition yields zero marginal DR against the planning hit, because the 50% cap is reached.

Glancing Blows and the Sequential Probability Chain

Glancing Blows is a keystone passive that inverts the luck of two defensive stats simultaneously: evasion rolls become *unlucky* (roll twice, take the worse) and deflection rolls become *lucky* (roll twice, take the better). For the Tactician, whose evasion is hard-capped at 50% by Stay Light and whose deflection is the primary win condition, Glancing Blows is the correct keystone — but it is important to understand what the tradeoff actually costs and pays.

Unlucky evasion at 50% cap:

$$P(\text{evade}) = P(\text{both rolls succeed}) = 0.50 \times 0.50 = 0.25$$

Lucky deflection at base chance d :

$$P(\text{deflect}) = 1 - (1 - d)^2 = 2d - d^2$$

At $d = 0.50$: $P(\text{deflect}) = 0.75$. At $d = 0.60$: $P(\text{deflect}) = 0.84$.

Glancing Blows trades 25 percentage points of evasion effectiveness ($50\% \rightarrow 25\%$ effective) for a large improvement in deflection rate. Whether this trade is favourable depends on the relative value of each avoidance layer, which the probability chain makes explicit.

Causal chain on an incoming hit:

1. Hit connects (accuracy check passes against evasion).
2. 25% chance to evade — hit is negated entirely. $P = 0.25$.
3. If not evaded (75%), deflection roll. At $d = 0.50$: 75% chance to deflect.
 $P(\text{not evade, deflect}) = 0.75 \times 0.75 = 0.5625$.
4. If not deflected (25% of non-evaded): armour alone. $P(\text{not evade, not deflect}) = 0.75 \times 0.25 = 0.1875$.

Table 23: **Expected Damage Taken at H=10,000 — Tactician (A=50,000, 25% Eff. Evasion, 75% Eff. Deflect)**

Outcome	Probability	DR	Damage Taken	Contribution
Evaded	25.0%	100%	0	0
Not evaded, deflected	56.25%	70.0%	3,000	1,688
Not evaded, not deflected	18.75%	50.0%	5,000	938
Expected	100%	73.8%	—	2,625

A=50,000 reaches the 50% DR cap on undeflected hits at H=10,000 ($2 \times 50k = 100k \geq 10 \times 10k$). Deflected hits take $0.6 \times 10,000$ at 50% DR = 3,000. Non-deflected hits take 10,000 at 50% DR = 5,000. The worst-case (non-evaded, non-deflected) requires 5,001 effective HP to survive — identical to Smith of Kitava's requirement at the same hit ceiling, but reached on a class whose gear and passive budget is split across armour, evasion, life, and resistances. Expected mitigation of 73.8% trails Shaman's passive 80.8% despite heavier defensive investment.

The survival implication of the worst-case row is the class's defining constraint: the 18.75% of hits that land without deflection require 5,001 effective HP. This is structurally identical to Smith of Kitava's requirement, but Tactician reaches this number on a mixed armour-evasion chassis that is competing for affixes across more defensive stat categories. Shaman, by contrast, requires only 1,923 HP to guarantee survival of the same hit — passively, with no probability chain.

The Ailment Threshold Benefit

Polish that Gear grants ailment threshold equal to 100% of evasion rating. Ailment threshold determines how much of a hit's damage must land before an ailment is applied. At high evasion investment (50,000–70,000 evasion rating), the threshold is large enough that:

- **Shock and freeze** — which require a hit to exceed a threshold proportional to maximum life — become functionally intermittent. At sufficient evasion

investment, most endgame hits fall below the threshold and these ailments apply so rarely they cease to be a defensive concern.

- **Ignite, bleed, and poison** — damage-over-time ailments not governed by threshold in the same way — still require charm-based removal.

This is not immunity; it is functional suppression of two of the most punishing ailments in endgame content (Shocked ground, freeze during a dangerous encounter) at the cost of evasion investment the build is already pursuing. The ailment suppression is a compounding secondary benefit, not the primary reason to invest in evasion.

Build Path Divergence: Evasion Hybrid vs. Block-and-Armour

The foregoing analysis assumes the standard Tactician build: evasion capped at 50% (unlucky to 25%), deflection from 20% of armour, and Stay Light, Use Cover providing 200% armour defend. A second path is viable — and in some respects more internally consistent — that forgoes evasion entirely:

Table 24: **Tactician Build Path Comparison**

Property	Evasion Hybrid (Standard)	Block-and-Armour (Alt)
Ascendancy nodes	Polish that Gear + Stay Light	Polish that Gear only
Avoidance layer	25% eff. evasion (unlucky 50%)	50% block
DR cap	50% (imposed by Stay Light)	None (uncapped armour DR)
Armour defend	200%	Standard (100%)
Armour target	$5 \times -10 \times L$	$10 \times -20 \times L$ (higher, no modifier)
Deflect source	20% of armour	20% of armour (same)
DR on large hits (H=10k)	50% hard cap	Uncapped; scales with armour invest
Ailment threshold	Yes (evasion-dependent)	Reduced (less evasion)
Gear budget pressure	High (armour, evasion, life, res)	Moderate (armour, life, res)
Win condition	Probabilistic: evade or deflect	Heavy deflect + high armour D

The case for the Block-and-Armour path: Forgoing Stay Light, Use Cover frees two ascendancy points, removes the 50% DR cap, and removes the evasion cap. Replacing evasion with block (50% block = 50% avoidance, not penalised by Glancing Blows) provides equivalent or superior avoidance compared to the unlucky 25% effective evasion the hybrid path achieves. With heavy armour investment, the deflection rate (lucky from Glancing Blows at 20% of armour) rises with armour — both deflect chance and raw armour DR increase simultaneously. If deflection applies front-loaded, a deflected hit at high armour values can achieve 60–70% armour DR

on the $0.6H$ remainder, producing a total effective DR of 76–82% on deflected hits — approaching Shaman’s passive 80.8%.

The unresolved constraint: Both paths depend on the same unknown: the actual deflection chance formula. If deflection rating of $0.2A$ at, say, $A = 100,000$ (deflect rating = 20,000) produces a high base chance, the lucky Glancing Blows roll yields a reliable 85%+ deflect rate, and the Block-and-Armour path is efficient. If deflection rating scales poorly or has a hard cap unrelated to armour, the premise collapses. This is an empirical question — not resolvable from the tooltip text alone — and represents the primary measurement gap in this section’s analysis.

The Identity Crisis: Optimisation Without a Dominant Constraint Solver

The Tactician’s core tension is this: the build imposes two independent scaling requirements (armour large enough to matter and evasion large enough for ailment threshold and avoidance) without providing a mechanism that converts one into the other at efficiency. The Invoker’s “...and protect me from harm” node converts evasion directly into armour. The Witchhunter’s Sorcery Ward converts armour into an absorption pool. The Tactician converts armour into deflection rating (20%) — a valuable secondary benefit, but one whose payoff depends on a probability distribution rather than a guaranteed quantity.

This produces the optimisation problem the user correctly identifies: you must plan armour for the worst-case (no deflect) and plan life pool for the worst-case (no evade, no deflect), but the build is rewarded most when both fire. The character who deflects consistently takes 3,000 from a 10,000 hit on 50,000 armour; the character who does not deflect takes 5,000 from the same hit and needs 5,001 effective HP to survive. These two planning targets do not converge.

The Witchhunter resolves this tension structurally: Sorcery Ward absorbs the post-armour remainder whether or not a deflection occurred, providing a deterministic secondary layer. The Tactician has no equivalent. Its survival of extreme hits is conditional on the probability chain resolving favourably, which is an inherent property of the archetype rather than a gap correctable through investment.

Tactician Positioning

Table 25: **Tactician vs. Shaman vs. Witchhunter at H=10,000**

Property	Shaman	Witchhunter	Tactician
Mitigation mechanism	Conversion (DR suppression)	SW absorption	Deflect + c
Expected damage (H=10k)	1,923 (passive)	0 (SW intact)	2,625 (prob
Worst-case damage	1,923 (deterministic)	5,000+ (SW depleted)	5,000 (no d
Min HP to survive worst case	1,924	3,001 (+ SW buffer)	5,001
Chaos coverage	Near-zero residual	SW absorbs post-DR	Armour DR
Armour target	Minimal (conversion does work)	$6 \times -10 \times L$ (+ SW)	$8 \times -10 \times L$ (
DR cap	None (conversion asymptote)	50% (post-penalty) + SW	50% hard c
Unique item required	Yes (Cloak of Flame)	No	No
Build expression delay	None	None	Post-campa
Verdict	Lowest HP floor, passive	Highest one-shot ceiling	Efficient a

Tactician expected mitigation (73.8%) trails Shaman (80.8%) despite heavier combined investment, because the 50% DR cap limits the ceiling on undeflected hits and the mixed gear requirements constrain life pool depth. Its structural advantage is armour efficiency: the 200% defend modifier halves the armour required to reach the DR cap versus a zero-conversion build, making the armour target genuinely achievable. The build is rewarding under the probability chain — deflect + DR produces 70% effective DR, comparable to Shaman — but cannot guarantee that chain fires on every hit, which is Shaman’s unconditional guarantee.

14 Warbringer: Block-as-Mitigation and the Mapping Specialist

Class Overview and Build Identity

The Warbringer is an ascendancy of the Marauder base class and represents an interesting adjacent case to Smith of Kitava: both are Marauder-lineage archetypes that lean into armour, block, and physical survivability, but the Warbringer resolves the armour formula’s scaling weakness through a different structural intervention. Rather than using conversion or absorption, the Warbringer converts block chance into a front-loaded damage multiplier — reducing hits that connect through its block roll to 20% of their original value before armour evaluates. Combined with a totem-linked damage reduction layer, the Warbringer achieves excellent *expected* mitigation across the hit-size spectrum. Its constraint is variance: the 25% of hits that bypass block land at 80% of their value and expose the same fundamental armour formula weakness that limits every non-converting archetype at large hit sizes.

The build’s natural home is totem-based content: high-uptime totems enable the secondary mitigation layer (Wooden Wall) near-continuously, and the ranged playstyle

reduces the frequency of hits that reach the character at all. Against boss-tier content with single large strikes, the probabilistic nature of block creates the defining tension — one the section explores in detail.

The Two Core Defensive Nodes

Table 26: Warbringer Ascendancy Nodes — Defensive Core

Node	Effect	Priority
Turtle Charm	Maximum block raised to 75%. You take 20% of the damage of blocked hits (80% less — confirmed front-loaded). Enables blocking of previously unblockable hits.	Mandatory
Wooden Wall	20% of damage you would take is instead taken by your totem. Requires an active totem.	Mandatory
Third/Fourth Choice	Offensive scaling, totem enhancement, or utility investment.	Situational

Turtle Charm's 80% less on blocked hits is confirmed to apply before the armour formula — the armour formula evaluates against 20% of the original hit, not the full hit. Wooden Wall's 20% reduction applies simultaneously, reducing the pre-armour value further. Both multipliers apply regardless of hit type (physical, elemental, chaos) provided the blocking condition and totem condition are met. Near-100% totem uptime is assumed throughout this analysis given the Warbringer's identity as a totem build.

Combined Multipliers: The Blocked-Hit Arithmetic

With both nodes active and a hit successfully blocked, two front-loaded multipliers apply in sequence:

$$\text{Turtle Charm (blocked): } \times 0.20 \quad \text{Wooden Wall (totem up): } \times 0.80$$

$$\text{Pre-armour effective hit (blocked): } = 0.20 \times 0.80 \times H = 0.16H \quad \Rightarrow \quad 84\% \text{ less damage before armour}$$

On an unblocked hit with Wooden Wall active:

$$\text{Pre-armour effective hit (unblocked): } = 0.80H \quad \Rightarrow \quad 20\% \text{ less damage before armour}$$

The implications for the armour formula are stark. Armour is highly efficient against

small hits (the formula’s numerator dominates when $H_{\text{eff}} \ll A$) and poor against large hits (the denominator grows with H). By collapsing a 10,000 hit to a 1,600 effective hit on a blocked roll, the Warbringer places armour in its most efficient operating regime 75% of the time — while the other 25% exposes armour to an 8,000 effective hit it cannot handle as efficiently.

Hit-Size Survival Table

The following table uses $A = 50,000$ armour with near-100% totem uptime (Wooden Wall always active). $P(\text{block}) = 0.75$.

Table 27: **Warbringer Damage Outcomes at $A=50,000$ (Turtle Charm + Wooden Wall Active)**

Hit	Eff. Hit (Blk)	Blk Dmg	Eff. Hit (No Blk)	No Blk Dmg	Expected	Eff. Mit.
500	80	1	400	30	8	98.3%
1,000	160	5	800	110	31	96.9%
2,000	320	19	1,600	388	111	94.4%
4,000	640	73	3,200	1,249	367	90.8%
8,000	1,280	261	6,400	3,593	1,094	86.3%
10,000	1,600	388	8,000	4,923	1,522	84.8%

“Eff. Hit” is the effective hit size entering the armour formula after front-loaded multipliers. “Blk Dmg” and “No Blk Dmg” are the final damage values on each outcome. Expected = $0.75 \times \text{Blk Dmg} + 0.25 \times \text{No Blk Dmg}$. At $H=10,000$, the expected mitigation of 84.8% exceeds Shaman’s passive 80.8% — a counterintuitive result driven by armour’s high efficiency against the 1,600 effective blocked hit. The 4,923 unblocked damage on the final row is the class-defining constraint: surviving this outcome requires 4,924 effective HP, more than double Shaman’s 1,924 HP requirement.

The expected value result — 84.8% mitigation surpassing Shaman’s deterministic 80.8% — is the build’s genuine appeal in mapping content where many hits arrive and the block distribution averages out. Against a single boss slam where one non-block is fatal, the expected value is irrelevant; the 25% non-block probability is the only number that matters.

Armour Targets: The eHP Constraint

Unlike other conversion or absorption archetypes in this paper where armour recedes as a binding constraint, the Warbringer’s planning target for worst-case survival is dictated jointly by armour and eHP. The armour required to guarantee survival of an unblocked $H = 10,000$ hit falls steeply as the life pool grows:

$$A^* = \frac{10 \times 0.8H \times (0.8H - L)}{L}$$

Table 28: **Armour Required to Survive Unblocked H=10,000 (Wooden Wall Active, by Life Pool)**

Life Pool (L)	Eff. Hit (0.8H)	Required Armour	Multiple of L
3,000	8,000	133,333	44.4×
4,000	8,000	80,000	20.0×
5,000	8,000	48,000	9.6×
6,000	8,000	26,667	4.4×
8,000	8,000	0	0

The armour requirement falls quadratically with life pool growth. At $L = 3,000$, the armour required (133k) is not achievable within normal gearing. At $L = 5,000$, 48k armour — achievable on a dedicated armour build — guarantees survival. At $L = 8,000$ (equal to the effective unblocked hit), zero armour is needed: the pool absorbs the hit regardless. This table is the formal justification for the user’s observation that the Warbringer’s core constraint is eHP, not armour: raising the pool from 3k to 5k reduces the armour requirement from impractical (133k) to standard (48k). The recommended build path is an armour/Energy Shield hybrid where ES fills the pool to the 5k–6k range, enabling armour targets that are achievable through normal investment.

The Unblockable Hit Interaction

Turtle Charm enables the Warbringer to block hits flagged as unblockable — including boss slams and telegraphed area attacks that every other archetype must dodge manually or absorb at full value. This is the build’s most uniquely powerful defensive property and has no equivalent in the other archetypes examined in this paper.

The practical evaluation is probabilistic. At 75% block on a boss slam of magnitude H :

- **75% of occurrences:** Hit is blocked. Pre-armour damage = $0.16H$. Armour handles a tiny effective hit efficiently. The slam is survived trivially.
- **25% of occurrences:** Hit is not blocked. Pre-armour damage = $0.80H$ (Wooden Wall only). This is a large effective hit and survival depends entirely on whether the life pool and armour together absorb it.

Against a 10,000 raw slam at the gear levels in the table above: the 25% non-block outcome deals 4,923 damage at $A=50k$ or 3,556 damage at $A=100k$. Whether these

values are survivable is entirely pool-dependent. On a 5,000 HP pool: a 4,923-damage outcome is survived by 77 HP at $A=50k$; a 3,556-damage outcome is survived comfortably at $A=100k$. On a 3,000 HP pool: neither is survivable.

The user’s framing is precise: “you are playing an RNG game where you hope that you block the slam.” The build offers a 3-in-4 success rate on the worst hits in the game — a genuine advantage over builds that must dodge manually and rely on player execution — but cannot guarantee survival without a sufficient eHP pool behind the 25% failure case.

Build Path: Standard Block vs. Turtle Charm

A Warbringer who skips Turtle Charm retains standard block mechanics (50% chance, full immunity on blocked hits) and saves two ascendancy points. With Wooden Wall and a terminal 10% less multiplier from another source:

- **$P(\text{block}) = 0.50$:** Full immunity — 0 damage taken.
- **$P(\text{no block}) = 0.50$:** $0.80H$ from Wooden Wall, then armour DR, then $\times 0.90$ terminal multiplier.
- **Expected ($A=50k$, $H=10k$):** $0.50 \times 0 + 0.50 \times 4,430 = 2,215$ — 77.9% expected mitigation.

Against Turtle Charm’s 1,522 expected damage (84.8% mitigation), the standard block path takes 46% more expected damage per hit at equivalent armour. The unblocked worst-case damage is identical (the armour formula sees the same $0.8H$), but the probability of triggering it is 50% rather than 25%. The trade-off for the two saved ascendancy points — which can go to offensive or totem-enhancing nodes — is a substantially higher expected damage intake per encounter and twice the probability of the worst-case outcome. Turtle Charm is the correct defensive choice when the ascendancy budget allows it.

Warbringer Positioning

Table 29: Warbringer vs. Smith vs. Shaman at H=10,000

Property	Shaman	Smith of Kitava	Warbringer (A=100k)
Mitigation mechanism	Conversion	Pure armour	Block + front-loaded mu
Expected damage	1,923 (deterministic)	5,000	1,055
Worst-case damage	1,923	5,000	3,556
Min HP (worst case)	1,924	5,001	3,557
Armour investment	Minimal (conversion floor)	100k+	100k+
Chaos coverage	Near-zero residual	Dedication to Kitava	Armour DR only
Blocks unblockable hits	No	No	Yes (Turtle Charm)
Totem dependency	No	No	Near-mandatory (Wooden)
Best content	General endgame	Crit/chaos maps	Mapping, totem content
Verdict	Lowest HP floor, passive	Crit/chaos specialist	Best expected value; I

Warbringer at A=100k achieves the best expected mitigation of the three armour-anchored builds (89.5% vs Shaman's 80.8% vs Smith's 50%). This result is counterintuitive but mathematically sound: armour is extraordinarily efficient against the 1,600 effective blocked hit (86.2% DR), dragging the expected damage well below Shaman's passive threshold. The cost is the 3,557 HP minimum for the worst-case unblocked hit — a requirement that Shaman never faces because its passive mitigation makes the worst case 1,923 regardless of block outcomes. The Warbringer is genuinely the best Marauder-lineage defensive archetype examined in this paper, but it does not match the investment efficiency of Shaman, Infernalist, or Witchhunter, which achieve superior or comparable expected mitigation at a fraction of the armour investment and without the life pool constraint imposed by the 25% unblocked outcome.

15 Conclusion

The hit-size spectrum analysis confirms and quantifies what the original insight identified: Shaman’s conversion stack is not one defensive strategy among many — it is a categorically more efficient approach than armour scaling across the endgame-relevant hit range, and the advantage compounds exactly where it matters most.

1. **Small hits are not the argument, and no one said they were.** At $H=100\text{--}500$, both Smith and Shaman achieve near-identical mitigation ($>95\%$ for both). Armour works extremely well here regardless of build. The hit-size tables confirm this rather than contradict it — the point is that Shaman is already winning at small hits with $4\times$ less armour investment, and the margin grows continuously from there.
2. **The divergence begins at the threshold where hits become dangerous.** Smith falls below 90% mitigation at $H\approx 1,100$. Shaman crosses the same threshold near $H\approx 2,600$. This is the window where incoming hits transition from “chip damage” to “survival-defining,” and Shaman maintains efficient mitigation for more than double the hit size.
3. **The 30.8 pp gap at $H=10,000$ translates directly into survival requirements.** Smith needs 5,001 effective HP to guarantee survival of the endgame ceiling hit. Shaman needs 1,923. A 2,000 HP pool — achievable without dedicated life investment — is sufficient for Shaman. The same 2,000 HP pool leaves Smith dead by 3,000 HP on the identical hit.
4. **Mind over Matter compounds the advantage asymmetrically.** Shaman’s low final damage (1,923) means a small mana pool (769) covers the MOM redirect. Smith’s high final damage (5,000) requires 2,000 mana to cover the redirect — on top of the 5,000+ life requirement. MOM multiplies Shaman’s already-strong position; it is a costly compromise for Smith.
5. **The mitigation ceiling cannot be reached by armour alone.** Matching Shaman’s 80.8% total mitigation through pure armour scaling requires 421,000 armour rating. Smith’s practical ceiling of 100,000 achieves 50.0% . The gap is not a matter of investment degree — it is a structural property of what the armour formula alone can achieve versus what conversion-augmented layering can achieve.

The Shaman’s configuration — Cloak of Flame, three ascendancy conversion notables,

one passive tree notable, and an elemental anointment — is among the most point-efficient passive physical mitigation archetypes in Path of Exile 2’s current state. It converts the armour formula’s greatest weakness (hit-size scaling) into its greatest asset (small residual physical hits) while simultaneously collapsing the converted elemental portion through resistance and armour-to-elemental layers. The result is a build that survives the largest hits in the game on a 2,000 HP pool while freeing the passive budget for damage that other archetypes cannot afford to spend.

The extended analysis in Sections 10–14 demonstrates that the Shaman’s approach is one of six distinct efficient defensive paradigms, not the unique solution. The Infernalist Full configuration (80% total conversion with Chaos Inoculation) achieves 85.3% mitigation at $H=10,000$ — surpassing the Shaman’s 80.8% — by pushing conversion one step further and eliminating the chaos component categorically. The Witchhunter (Section 12) reaches a survival ceiling of approximately $H = 14,485$ at mid-endgame investment through Sorcery Ward absorption, which absorbs the full post-armour remainder of a 10,000 raw hit with zero life damage — without a unique-item dependency. The Invoker operates on an entirely separate axis: 75–88% of hits are avoided before any mitigation applies, and the hits that do land are handled by an evasion-derived armour pool and a pre-combat Energy Shield overflow via Meditate. The Tactician (Section 13) introduces a fifth paradigm: deflection as a front-loaded multiplier that reduces the effective hit entering the armour formula, combined with a 200% armour defend modifier that halves the armour investment required to reach the DR cap. Its expected mitigation of 73.8% at $H=10,000$ trails Shaman’s deterministic 80.8%, and its worst-case survival requirement (5,001 HP) is structurally identical to Smith of Kitava’s — but it achieves this on an armour target of $8-10 \times L$ rather than $20 \times L$, and its probability chain rewards both evasion avoidance and deflection simultaneously. The Warbringer (Section 14) adds a sixth paradigm: block converted into a front-loaded 80% damage multiplier, producing the highest expected mitigation of any armour-anchored build (89.5% at $A=100k$) at the cost of high variance on unblocked hits and a totem-uptime dependency for the secondary layer. These six archetypes — Shaman, Infernalist, Witchhunter, Tactician, Warbringer, and Invoker — collectively demonstrate that Path of Exile 2’s most efficient defensive builds share a common structural property: they reduce, convert, absorb, deflect, or avoid the effective damage rather than scaling raw armour to meet it. Each answers a different threat profile, and each is viable at high endgame investment.